BOOST LED driver: system schematic

Two boards, two KiCad projects. Open the power board with BOOST_power_RC2.kicad_pro and the control card with BOOST_control_RC2.kicad_pro:
each board project opens its own sheet (below) as its schematic, and its PCB updates from it. This root sheet ties both together for whole-system ERC.

Power board



File: BOOST_power_RC2.kicad_sch

Control card

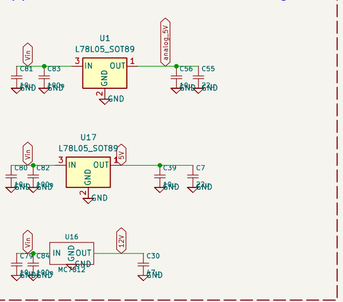


File: BOOST_control_RC2.kicad_sch

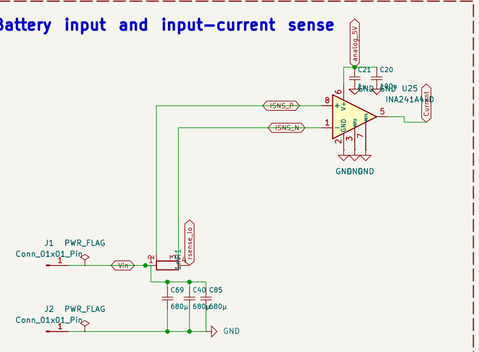
The boards connect through the 30-pin J10 (power board) / J11 (control card) pair: every signal that crosses between them carries the same global label on both sheets.
LTspice: LTspice/BOOST.asc simulates both boards together, laid out in the same blocks as these sheets (power board left, control card right, test bench below).

Sheet: /		
File: BOOST.kicad_sch		
Title: BOOST LED driver – system		
Size: A4	Date: 2026-10-05	**Rev: RC2**
KiCad E.D.A. 10.0.0		Id: 1/3

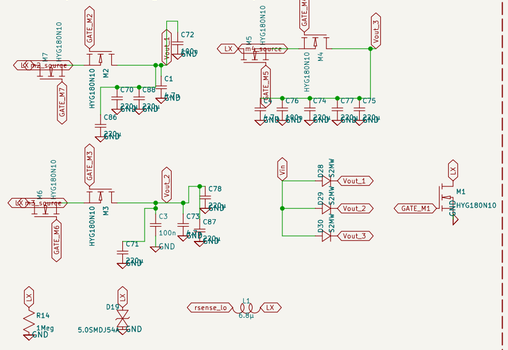
Supplies: 12 V and two 5 V regulators



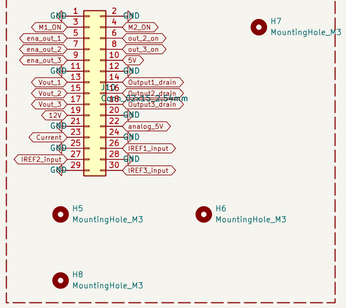
Battery input and input-current sense



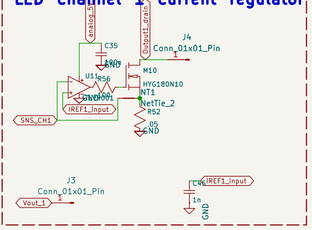
Boost power stage and output banks



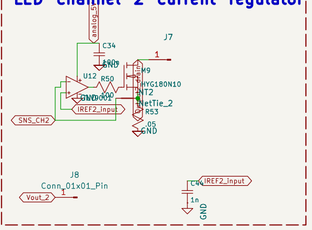
Card connector and mounting holes



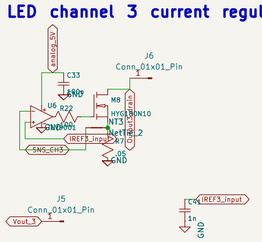
LED channel 1 current regulator



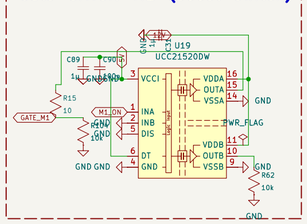
LED channel 2 current regulator



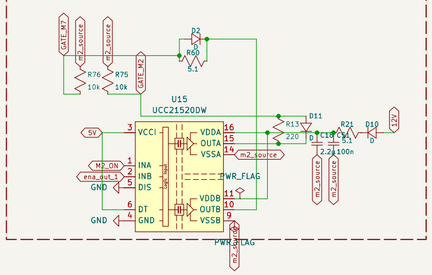
LED channel 3 current regulator



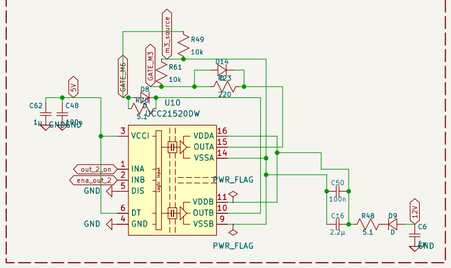
Gate driver: M1 (boost switch)



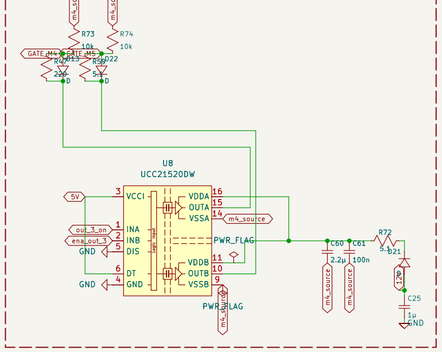
Gate driver: channel 1 switches M2/M7



Gate driver: channel 2 switches M3/M6



Gate driver: channel 3 switches M4/M5



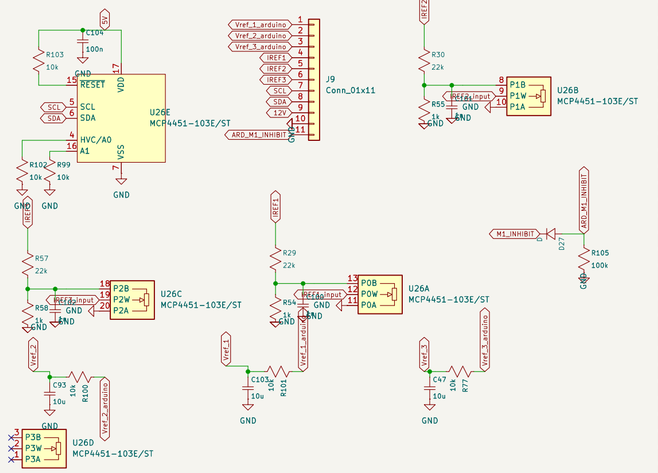
Nets between blocks are joined by global labels of the same name
One block per circuit: the LTspice schematic uses the same arrangement

Sheet: /Power board/
File: BOOST_power_RC2.kicad_sch

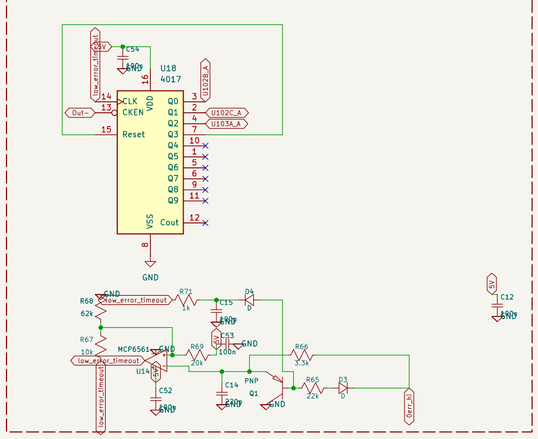
Title: BOOST LED driver – Power board

Size: A2	Date: 2026-10-05	**Rev: RC2**
KiCad E.D.A. 10.0.0		Id: 2/3

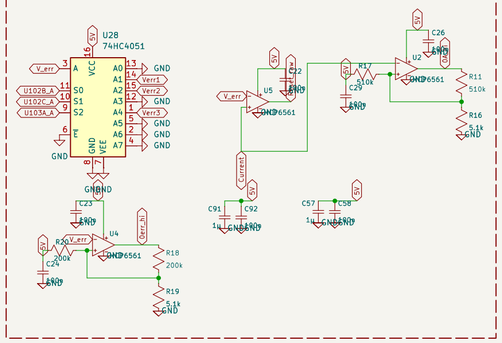
Arduino interface and LED-current references



Channel sequencer and low-error timeout



Error select and comparators



Channel 1 output-voltage error amplifier

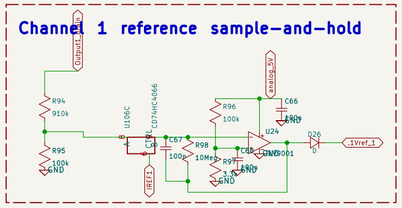


Channel 2 output-voltage error amplifier

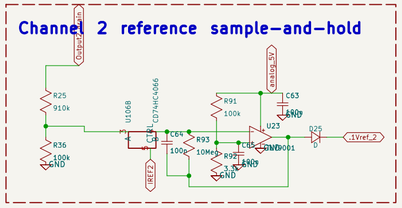


Channel 3 output-voltage error amplifier

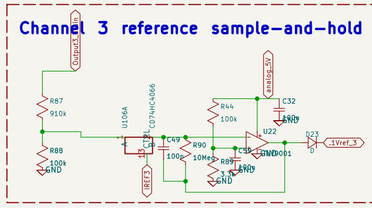
Channel 1 reference sample-and-hold



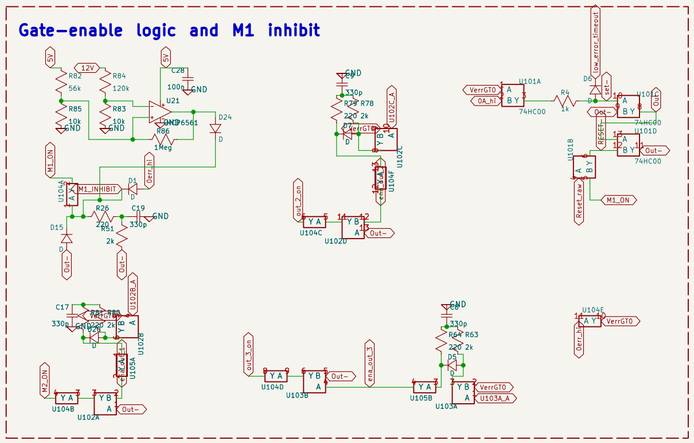
Channel 2 reference sample-and-hold



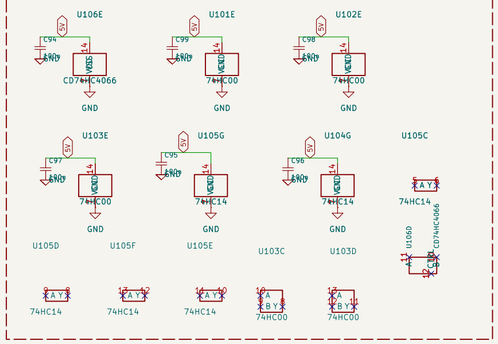
Channel 3 reference sample-and-hold



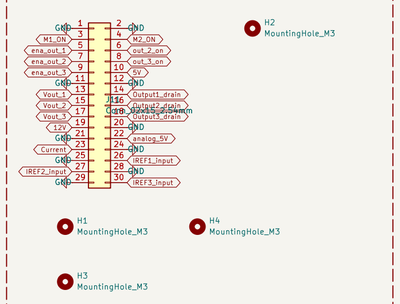
Gate-enable logic and M1 inhibit



Logic supplies and spare gates



Power-board connector and mounting holes



Nets between blocks are joined by global labels of the same name
One block per circuit; the LTspice schematic uses the same arrangement

Sheet: /Control card/		
File: BOOST_control_RC2.kicad_sch		
Title: BOOST LED driver – Control card		
Size: A2	Date: 2026-10-05	**Rev: RC2**
KiCad E.D.A. 10.0.0		Id: 3/3